Repeated Transaction Detection (Case Study 28)

Problem Statement

Check a list of transaction IDs and identify IDs that appear more than once.

Objectives

Practise lists, nested loops, and duplicate checking.

Assumptions

Transaction IDs are strings. The program reports each repeated ID only once.

Algorithm

1. Read IDs separated by spaces. 2. Create an empty duplicate list. 3. Check each ID's count. 4. Add repeated IDs if not
already reported. 5. Print the result.

Flowchart (Text Form)

START → Read transaction IDs → Check each ID count → Repeated and not reported? add to duplicates → Print
duplicates → STOP.

Pseudocode

Python Source Code

```
ids = input("Enter transaction IDs separated by spaces: ").split()
duplicates = []
for item in ids:
    if ids.count(item) &gt; 1 and item not in duplicates:
        duplicates.append(item)
if duplicates:
    print("Repeated transaction IDs:", duplicates)
else:
    print("No repeated transaction IDs")
```

Test Case and Expected Output

Input: T101 T102 T101 T103 T102
Expected output: Repeated transaction IDs: ['T101', 'T102']

Conclusion

The program detects repeated transaction identifiers using basic list operations.